

Plan: Delete MBR, unify on the OOM two-round scoring path

Pioneer.jl — branch feat/shadow-global-max-improvements

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1 Context and goal

Two-round cross-run scoring (per-fold run clustering → GROUP features → shadow-decoys → round-2 Light-GBM) plus Dennis's learned global model now supersede MBR for controlling false transfer while gaining IDs. The remaining work is to make two-round the *single* scoring path and remove MBR.

Phase 1 (done, 5b2f60418). `_write_group_features!` was rewritten to stream (bounded memory, $O(\text{vocab}) + \leq 9$ files, independent of run count) after the 208-file APMS Astral regression job hung ~10 h on the old all-in-memory version. Verified *bit-identical* to the prior output on EWZ.

Design decision (this doc). The `match_between_runs` config flag is repurposed to mean “do a second round”:

- `match_between_runs = true` → round-1 **and** round-2 (GROUP features + shadow-decoys).
- `match_between_runs = false` → round-1 only (first-round scores; no GROUP features, no shadow-decoys, no second training pass).

Both paths use the streamed OOM trainer. The old MBR machinery (counterfactual partner pools, donor dict, per-file MBR feature sidecars, paired-FTR controller, recovery sidecars) is deleted outright.

2 Current structure (before)

`score_precursor_isotope_traces` branches on the flag into two whole pipelines:

```
function score_precursor_isotope_traces(second_pass_folder, file_paths, precursors,
                                       fragment_lookup, ::Int64, ::Float32=0.01f0, ::Bool=false;
                                       match_between_runs::Bool = true)

    if match_between_runs
        return _score_precursor_isotope_traces_mbr(file_paths, precursors, fragment_lookup) # DELETE
    else
        return _score_precursor_isotope_traces_no_mbr(second_pass_folder, file_paths, precursors)
    end
end
```

The MBR path (`_score_precursor_isotope_traces_mbr`) runs: counterfactual pools → Pass-1 OOM → donor dict → per-file MBR feature sidecars → slim FTR DataFrame → paired-FTR filter → recovery sidecars → merge. The no-MBR path already uses the OOM trainer and, gated by the current forced `two_round_enabled()=true` hack, runs the two-round steps.

Forced overrides to remove (added for regression testing):

```
# PrecursorScoringSearch.jl -- forced MBR off
mbr = false # was: Bool(global_params.match_between_runs) : true
# two_round_scoring.jl -- forced two-round on
two_round_enabled() = true # was: get(ENV, "TWO_ROUND", "") == "1"
```

3 Phase 2 — Unify on one OOM path

Collapse the branch into a single function. `match_between_runs` now gates *only* the second round. `fragment_lookup` (MBR empirical-spectrum features) becomes unused and is dropped.

```

# score_psms.jl (after)
function score_precursor_isotope_traces(second_pass_folder, file_paths, precursors,
                                       ::Int64, ::Float32=0.01f0, ::Bool=false;
                                       match_between_runs::Bool = true)

    # Round 1: streamed OOM LightGBM over the per-file Arrow tables -> .pass1_sidecar.arrow (OOFS1).
    features = copy(ADVANCED_FEATURE_SET)
    pass1 = train_and_predict_pass1_oom!(file_paths; features=features,
                                         compute_infold=false, lgbm_hp=SCORING_LGBM_HP, semisupervised=true)
    log_pass_importance(pass1, match_between_runs ? "Round-1" : "Pass-1")

    if match_between_runs
        # Round 2: cross-run GROUP features + shadow-decoys, re-trained on the same streamed OOM path.
        write_two_round_feature_columns!(file_paths, length(precursors)) # streaming (Phase 1)
        inject_shadow_decoys!(file_paths)
        features2 = vcat(copy(ADVANCED_FEATURE_SET), two_round_features())
        pass1 = train_and_predict_pass1_oom!(file_paths; features=features2,
                                             compute_infold=false, lgbm_hp=SCORING_LGBM_HP, semisupervised=true)
        log_pass_importance(pass1, "Round-2 (+GROUP cross-run,+delta_irt)")
        log_shadow_fdr_diagnostics(file_paths; q_threshold=0.01)
        remove_shadow_decoys!(file_paths)
    end

    _merge_pass1_into_main!(file_paths, precursors) # renamed from _merge_pass1_into_main_no_mbr!
    return nothing
end

```

`two_round_enabled()` is deleted; the `match_between_runs` argument is the single gate. Its call site (`PrecursorScoringSearch.jl`) already passes `params.match_between_runs` — that stays, but now sourced from config again (Phase 4).

4 Phase 3 — Delete MBR machinery

Delete functions (all only reachable from the removed MBR path): `_score_precursor_isotope_traces_mbr`, `build_counterfactual_partner_pools`, `build_mbr_fragment_annotation_keys`, `build_mbr_donor_dict_streaming_with_params`, `compute_mbr_features_per_file_to_sidecar_with_pass1!`, `load_ftr_slim_dataframe`, `apply_mbr_filter_paired!`, `write_recovery_sidecars`, `merge_mbr_sidecars_into_main!`.

Delete files and their `importScripts.jl` entries:

```

"mbr_pairing.jl",      # MBR Phase 1: 1:1 pair regeneration with cloning
"mbr_streaming.jl",    # MBR Phase 2: donor dict + per-file MBR sidecars
"mbr_ftr.jl",          # MBR Phase 4: FTR controller on MBR-boosted score

```

Downstream compatibility (verified graceful). Protein inference / MaxLFQ read MBR outputs only behind `hasproperty` guards, so their absence takes the default branch:

```

# protein_inference_pipeline.jl -- already safe when the column is gone
if !hasproperty(gdf, :mbr_recovered)
    return (recovered_peptides = Int64(0), only_protein = false) # default: nothing recovered
end
# MBR_boosted_prec_prob / MBR_boosted_qual / MBR_boosted_global_qual: all hasproperty-guarded ->
# fall through to the plain :prec_prob / :qual / :global_qual columns.

```

The no-MBR merge currently writes `mbr_recovered = false`. Two options: (a) keep writing the constant-`false` column for schema stability, or (b) stop writing it and rely on the `hasproperty` defaults. **Choose (a)** — one cheap column, avoids touching the MBR-boosted-column selection logic and the `mbr_qual_bypass` step in `PrecursorScoringSearch.jl`, and keeps output schema stable for regression baselines.

`MainSearch` computes *no* MBR features (verified), so nothing upstream needs changing.

5 Phase 4 — Remove forced overrides; config-drive the flag

```

# PrecursorScoringSearch.jl -- restore config read (default true = do second round)
mbr = hasproperty(global_params, :match_between_runs) ?
    Bool(global_params.match_between_runs) : true
# two_round_scoring.jl -- delete two_round_enabled() entirely (match_between_runs is the gate)

```

Net effect: `match_between_runs` in the config now selects one-round vs two-round scoring, with no MBR machinery behind either.

6 Phase 5 — Validation

1. **Bit/ID-equivalence:** EWZ two-round (`match_between_runs=true`) unchanged vs the current 5b2f60418 two-round output (deletion must not alter the two-round path).
2. **One-round path:** EWZ with `match_between_runs=false` produces round-1-only scores (no GROUP columns, no shadows) and completes.
3. **Scale:** APMS_Astral (208 files, `match_between_runs=true`) completes in reasonable time/memory (the original hang).
4. **Full regression** on the branch; confirm entrapment FDR unchanged (precursor + experiment-wide protein calibration from the earlier entrapment work).

7 Risk and rollback

Large deletion, but low-risk because: (i) two-round path is untouched by the deletion (only the MBR branch and dead code go); (ii) downstream MBR reads are `hasproperty`-guarded; (iii) Phase 1 already proved the two-round path scales. Pre-change state is preserved on `origin` (`backup/shadow-global-max-pre-rebase`) and each phase is a separate commit, so any phase is individually revertible. Git history retains the deleted MBR code if it is ever needed again.